

SecureShift Backend Onboarding

Windows 11 + WSL2 and macOS | Docker, host-run backend, MongoDB Compass, seed data and email

Version 1.1 — updated for local development seed data

Recommended everyday backend setup: Run MongoDB in Docker, run the backend directly in WSL or macOS with Nodemon, and connect MongoDB Compass to localhost.

Recommended full-system check: Run MongoDB, backend and employer frontend together with Docker Compose.

1. Understand the two MongoDB hostnames

- Use mongodb:27017 only when the backend itself runs inside Docker Compose.
- Use localhost:27017 when the backend runs directly in WSL, Windows or macOS.
- MongoDB Compass on Windows or macOS also connects to localhost:27017.

Inside Docker Compose:

```
mongodb://seureshift_app:seureshift_app_password@mongodb:27017/seureshift_local?authSource=seureshift_local
```

Outside Docker / Compass:

```
mongodb://seureshift_app:seureshift_app_password@localhost:27017/seureshift_local?authSource=seureshift_local
```

2. First-time setup after pulling the Docker onboarding changes

Warning: The reset below permanently deletes the existing local Docker MongoDB volume. Existing developers need it once after the database name and credentials change. Fresh clones do not need it.

```
cd ~/developer/SecureShift
git switch main
git pull --ff-only
docker compose down -v
docker compose up --build -d
docker compose ps
```

After this one-time reset, use docker compose down without -v for normal shutdown.

3. Full Docker Compose workflow

Windows / WSL2

```
cd ~/developer/SecureShift
docker compose up --build -d
docker compose ps
curl -i http://localhost:5000/api/v1/health
```

- Employer panel: <http://localhost:3000>
- Swagger: <http://localhost:5000/api-docs/>
- MongoDB Compass: use the localhost connection in Section 7.

macOS

If port 5000 is occupied, create a root .env file:

```
cd /path/to/SecureShift
cp .env.example .env
```

Uncomment or add:

```
BACKEND_HOST_PORT=5001
```

```
docker compose up --build -d
docker compose ps
curl -i http://localhost:5001/api/v1/health
```

- Employer panel: <http://localhost:3000>
- Swagger: <http://localhost:5001/api-docs/>
- MongoDB Compass still uses localhost:27017 unless MONGODB_HOST_PORT is overridden.

4. Recommended backend-development workflow

This is the fastest workflow for backend contributors. MongoDB stays in Docker, while the backend runs directly with Nodemon for immediate reloads and easier debugging.

Windows / WSL2

```
cd ~/developer/SecureShift
docker compose down
docker compose up -d mongodb
docker compose ps
```

Create or edit app-backend/.env:

```
NODE_ENV=development
PORT=5000
MONGO_URI=mongodb://seureshift_app:seureshift_app_password@localhost:27017/seureshift_local?authSource=seureshi
ft_local

JWT_SECRET=<generated-private-secret>
LICENCE_ENC_KEY=<generated-private-key>

EMAIL_ENABLED=false
SMTP_HOST=localhost
SMTP_PORT=587
SMTP_SECURE=false
SMTP_USER=local@example.test
SMTP_PASS=local-dev-password
SMTP_FROM_EMAIL=local@example.test

SEED_ALLOW_LOCAL=false
SEED_RESET_CONFIRM=

cd app-backend
git check-ignore -v .env
npm install
npm run dev

curl -i http://localhost:5000/api/v1/health
```

macOS

```
cd /path/to/SecureShift
docker compose down
docker compose up -d mongodb
docker compose ps
```

Create or edit app-backend/.env. If port 5000 is occupied, use PORT=5001:

```
NODE_ENV=development
PORT=5001
MONGO_URI=mongodb://seureshift_app:seureshift_app_password@localhost:27017/seureshift_local?authSource=seureshi
ft_local

JWT_SECRET=<generated-private-secret>
LICENCE_ENC_KEY=<generated-private-key>

EMAIL_ENABLED=false
SMTP_HOST=localhost
SMTP_PORT=587
```

```
SMTP_SECURE=false
SMTP_USER=local@example.test
SMTP_PASS=local-dev-password
SMTP_FROM_EMAIL=local@example.test
```

```
SEED_ALLOW_LOCAL=false
SEED_RESET_CONFIRM=
```

```
cd app-backend
git check-ignore -v .env
npm install
npm run dev
```

```
curl -i http://localhost:5001/api/v1/health
```

5. Generate local backend secrets

Run these commands separately on each machine:

```
openssl rand -hex 32
openssl rand -base64 32
```

- Paste the first value into JWT_SECRET.
- Paste the second value into LICENCE_ENC_KEY.
- Different values on WSL and Mac are fine for independent local databases.
- Use the same values only when both machines must decrypt the same encrypted records or preserve compatible sessions against a shared database.
- Keep each LICENCE_ENC_KEY stable once meaningful local encrypted data exists.
- Never commit app-backend/.env.

6. Populate the local database with seed data

The seed workflow creates deterministic fictional development data for backend, employer frontend and Guard App testing. It is idempotent: running it again updates the same records rather than duplicating them.

Safety: The seed refuses production, Atlas/SRV, remote hosts and any database except secureshift_local, secureshift_dev or secureshift_test. Reset deletes only deterministic seed IDs and requires an exact confirmation string.

Backend outside Docker; MongoDB in Docker

In app-backend/.env, temporarily set:

```
NODE_ENV=development
SEED_ALLOW_LOCAL=true
MONGO_URI=mongodb://secureshift_app:secureshift_app_password@localhost:27017/secureshift_local?authSource=secureshift_local
```

Then run from app-backend:

```
npm run seed
SEED_RESET_CONFIRM=SecureShiftLocalReset npm run seed:reset
```

Reset: seed:reset removes only known seeded records. It does not empty the entire database. Set SEED_ALLOW_LOCAL back to false when you do not need seeding.

Seed alongside Docker Compose

From the repository root, use one-off backend containers. Compose supplies the internal mongodb hostname automatically:

```
docker compose run --rm -e SEED_ALLOW_LOCAL=true backend npm run seed
```

```
docker compose run --rm -e SEED_ALLOW_LOCAL=true -e SEED_RESET_CONFIRM=SecureShiftLocalReset backend npm run seed:reset
```

Seeded test accounts

All seeded accounts use the local-only password SecureShift1!. Employer and guard login still use the normal OTP flow; the seed does not bypass OTP.

Role	Scenario	Email
Admin	Admin access	admin.local@seureshift.test
Employer	Operations employer	ops.local@seureshift.test
Employer	Venue employer	venue.local@seureshift.test
Guard	Approved licence	mia.guard@seureshift.test
Guard	Pending licence	noah.guard@seureshift.test
Guard	Rejected licence	isha.guard@seureshift.test
Guard	Expired licence	liam.guard@seureshift.test

The seed also creates roles, two branches, guard availability, five shift scenarios, attendance, isolated payroll, messages and notifications. Payroll generation is scoped to the deterministic seeded completed shift and does not absorb unrelated developer-created shifts.

Fixture time zones: Some seed helpers currently mix local-time and UTC calculations. Scenarios and durations remain correct, but absolute stored timestamps can differ between Docker UTC and a host such as Adelaide. This is acceptable for local fixtures and may be normalised later.

7. MongoDB Compass connection

Create a new Compass connection named SecureShift Local Docker:

```
mongodb://seureshift_app:seureshift_app_password@localhost:27017/seureshift_local?authSource=seureshift_local
```

If localhost does not work, try:

```
mongodb://seureshift_app:seureshift_app_password@127.0.0.1:27017/seureshift_local?authSource=seureshift_local
```

Confirm MongoDB is published to the host:

```
docker compose ps
```

Expected port mapping:

```
0.0.0.0:27017->27017/tcp
```

Check collection document counts:

```
docker compose exec mongodb mongosh
"mongodb://seureshift_app:seureshift_app_password@localhost:27017/seureshift_local?authSource=seureshift_local"
--quiet --eval 'db.getCollectionNames().forEach(c => print(c + ": " + db.getCollection(c).countDocuments({})))'
```

Empty collections: After docker compose down -v, previous local documents are gone. Empty collections with indexes are normal until data is created or npm run seed is run.

8. MongoDB 6 end-of-life warning

MongoDB Compass may warn that MongoDB 6 has reached end of life because the current Compose file uses mongo:6. This does not prevent local development and is not the cause of empty collections.

- Do not silently change docker-compose.yml on main.
- Treat a MongoDB image upgrade as a separate tested PR.
- Test application compatibility, clean-volume startup, authentication, health checks and the migration path.

9. Email and verification codes

EMAIL_ENABLED=false means no genuine SMTP account is required. The local@example.test values are placeholders; real emails and verification codes will not be sent.

To test real email delivery, obtain the team's private SMTP configuration:

```
EMAIL_ENABLED=true
SMTP_HOST=<real SMTP host>
SMTP_PORT=<provider port>
SMTP_SECURE=<true or false>
SMTP_USER=<real testing account>
SMTP_PASS=<SMTP password or app password>
SMTP_FROM_EMAIL=<approved sender address>
```

- Store real credentials only in the ignored app-backend/.env.
- Never post SMTP passwords in GitHub, Teams channels or documentation.
- The seed does not bypass OTP. When email is disabled, use Compass or an approved development workflow to inspect local users and statuses.

10. Validation checklist

- docker compose ps shows secureshift-db healthy.
- Backend health returns HTTP 200 with {"status":"ok"}.
- Compass connects to secureshift_local through localhost:27017.
- npm run seed produces stable counts on repeated runs.
- npm run seed:reset removes seeded records but preserves unrelated local records.
- git check-ignore -v app-backend/.env confirms the private environment file is ignored.
- No real passwords, SMTP credentials, JWT secrets or encryption keys are committed.

11. Troubleshooting

Symptom	Likely cause and fix
getaddrinfo EAI_AGAIN mongodb	Backend is outside Docker. Change @mongodb:27017 to @localhost:27017 in app-backend/.env.
Compass connects but shows zero documents	The volume was reset, this is a different local database, or seed data has not been loaded. Run npm run seed or the Docker seed command.
Mongo healthcheck fails with UserNotFound	An old volume still contains old credentials. Run docker compose down -v once, then rebuild.
Port 5000 already in use on Mac	Use PORT=5001 for a host-run backend and BACKEND_HOST_PORT=5001 for the full Compose stack.
Seed refuses to run	Confirm NODE_ENV is not production, SEED_ALLOW_LOCAL=true, the URI is explicit and authenticated, and the database name is exactly secureshift_local, secureshift_dev or secureshift_test.
Seed reset refuses to run	Set SEED_RESET_CONFIRM=SecureShiftLocalReset for that command.
No verification email arrives	EMAIL_ENABLED is false or SMTP settings are placeholders. Use valid private SMTP settings.
Atlas URI returns ESERVFAIL	This is separate from local Docker MongoDB. Use the localhost connection for local work.

12. Daily command summary

Backend development - WSL

```
cd ~/developer/SecureShift
docker compose up -d mongodb
cd app-backend
npm run dev
```

Backend development - Mac

```
cd /path/to/SecureShift
docker compose up -d mongodb
cd app-backend
npm run dev
```

Load local seed data - host-run backend

```
cd app-backend
npm run seed
```

Load local seed data - Docker

```
docker compose run --rm -e SEED_ALLOW_LOCAL=true backend npm run seed
```

Full stack check

```
cd /path/to/SecureShift
docker compose down
docker compose up --build -d
docker compose ps
```

Normal shutdown

```
docker compose down
```